



Digital Twin of Portuguese Wildfires

Group – N

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Digital Shadow



Wildfire Events



AI Prediction



Our Goal

Creating a digital twin to analyze historical wildfire data in Portugal, identifying high-risk areas and predicting future events.



Digital Twin

Creating a digital shadow architecture to model wildfire events



Data Analysis

Analyzing historical Portuguese wildfire data from 2013-2022



Risk Identification

Identifying high-risk areas prone to future wildfire events



Future Prediction

Predicting future events to help authorities allocate resources effectively

How It Helps

Supporting Civil Protection authorities with three key capabilities



Understanding Past

Replaying and visualizing historical wildfire seasons to understand patterns and trends over time

- ✓ Provides historical context



Simulating Events

Modeling real-time wildfire report flows to understand system dynamics and improve response strategies

- ✓ Enhances system comprehension



Predicting Future

Using AI to predict high-risk areas, helping authorities allocate resources effectively

- ✓ Optimizes resource allocation

Data Sources

Portugal wildfire event dataset (2013-2022) from ICNF (Instituto de Conservação da Natureza e das Florestas)



Location Details

Latitude, longitude, and district information for each wildfire event



Event Details

Alert timestamp, first intervention timestamp, and extinction timestamp for each event



Environmental Factors

Temperature, humidity, wind speed, precipitation, and fire weather index (FWI)



Geography and Impact

Burned total area, including population, brushland, and agriculture affected

Dataset Example

Sample data from the Portuguese wildfire dataset (2013-2022) provided by ICNF

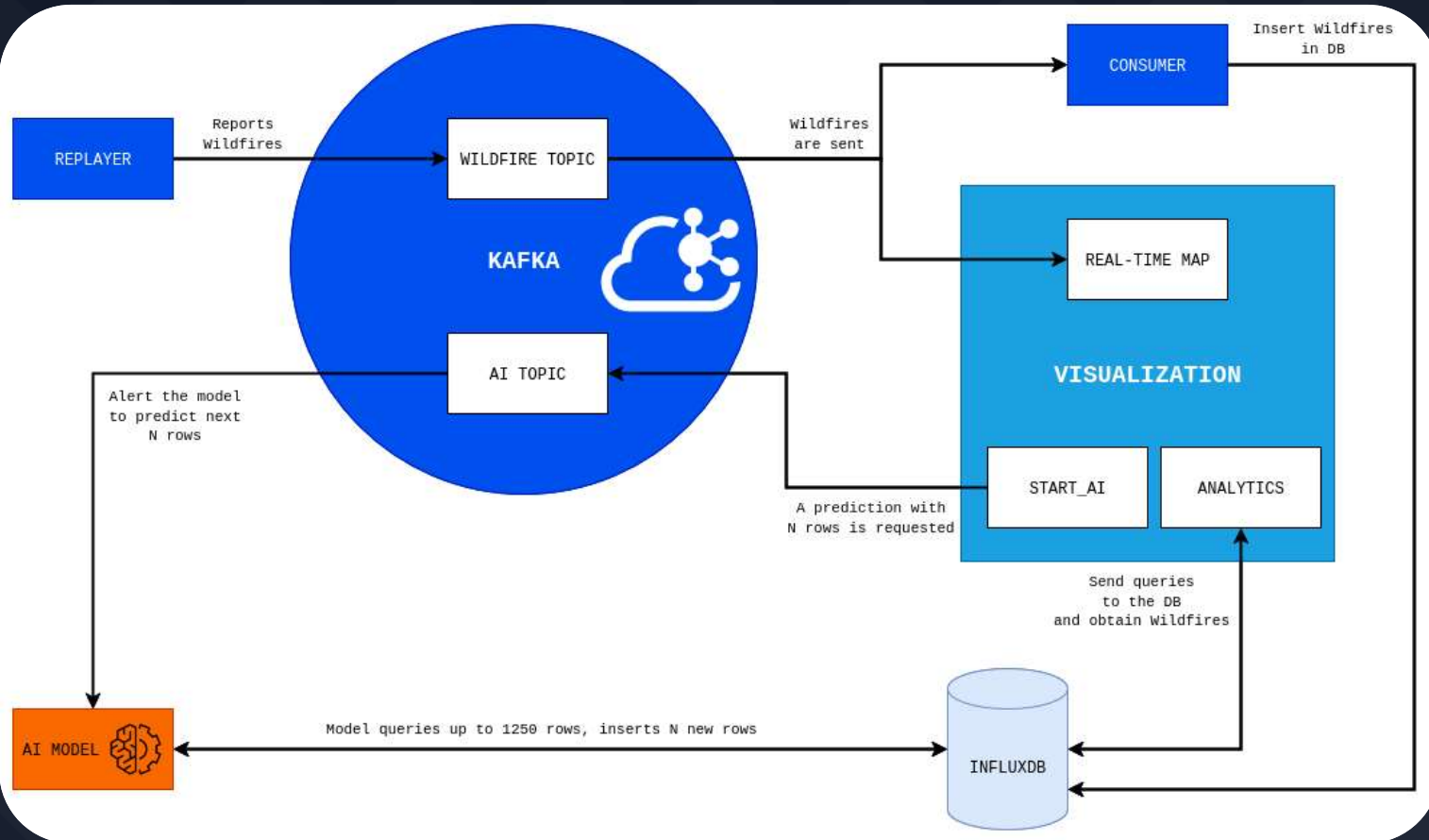
 Timestamp	 Region	 Wildfire Type	 Burned Area	 Temperature	 Humidity	 Wind Speed
1357235640	Coimbra	Wildfire_Small	0.001001714	14.60°C	41%	10 m/s
1357305180	Leiria	Wildfire_Small	39.8961	9.60°C	51%	28 m/s
1357410000	Leia	Agricultural	0.07	14.89°C	42%	14 m/s
1357574640	Faro	Wildfire_Small	0.235294	16.39°C	73%	0 m/s
1357673400	Coimbra	Wildfire_Small	0.002	13.80°C	68%	68 m/s

-  Timestamp & Location
-  Wildfire Type & Area
-  Environmental Factors
-  Geographic Impact

Digital Shadow Architecture

Digital Shadow Architecture

Our system adopts a "Digital Shadow Architecture" that models wildfire events as digital events, enabling analysis and prediction.



Data Replay

Replaying historical wildfire data



Messaging

Kafka broker for real-time data flow



Prediction

AI model for future event prediction



Processing & Storage

Analytics and InfluxDB storage

The Entity: Wildfire Event

Core Concept



Wildfire Event

The system is built around the core concept of "Wildfire Event," using a hybrid data-driven representation model that combines multiple information sources.

"A Wildfire Event is represented as a complex entity with multiple attributes spanning from location and timeline to environmental context."

Hybrid Data-Driven Model

Combines data-driven learning with physical information features to create a comprehensive representation of wildfire events.



Data-Driven Learning

Learning patterns from historical wildfire data to identify trends and predict future events.



Physical Information

Incorporating terrain and environmental data to contextualize wildfire behavior.

Key Attributes

Essential variables for modeling wildfire events



Location

- LATITUDE: Geographic latitude coordinate
- LONGITUDE: Geographic longitude coordinate



Timeline

- ALERT_TIMESTAMP: Time of initial alert
- FIRST_INTERVENTION_TIMESTAMP: Time of first response
- EXTINCTION_TIMESTAMP: Time when fire was extinguished



Impact

- BURNED_TOTAL_AREA: Total area affected by the wildfire
- BURNED_POPULATION: Population density in affected areas



Weather

- TEMPERATURE_CELSIUS: Ambient temperature
- FWI: Fire Weather Index
- RELATIVE_HUMIDITY: Percentage of humidity

Prediction Model

Microsoft LightGBM model for forecasting future wildfire event characteristics



Model Goal

Forecasting future wildfire characteristics



Features

Uses historical patterns to predict future events



System Impact

Enriches the Digital Twin from historical to predictive tool

Why LightGBM?

Advantages of using LightGBM for this project



Fast

High performance on CPU

Crucial for a responsive system

"LightGBM's efficient CPU performance makes it ideal for real-time wildfire prediction."



Accurate

Powerful gradient boosting model

Well-suited for tabular data

"LightGBM's accuracy enables precise prediction of wildfire characteristics."



Resilient

Handles irregular timestamps

Where other time-series models struggled

"Perfect for handling the irregular data patterns in our wildfire dataset."

"The AI enriches our Digital Twin, elevating it from a historical replay system to a predictive tool."

Interactive Dashboard

Our system provides insights through a custom-built dashboard that allows users to visualize events, analyze metrics, and generate forecasts.



Interactive Map

Track fires in simulated real-time with pan, zoom, and click-for-details functionality.

Visualize hotspots



Statistical KPIs

View key metrics including total fires, area burned, and trends over time.

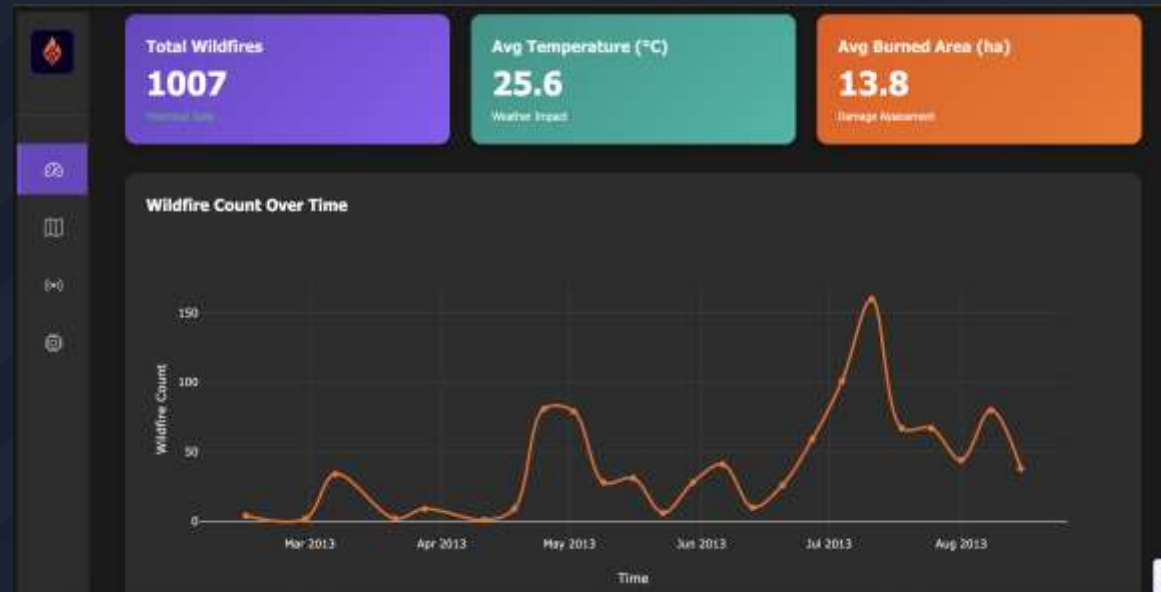
Analyze trends



AI-Driven Forecasts

Generate AI predictions for future events and visualize them on the map to support planning.

Predict & plan





Digital Twin in Action

Transition to your live demo of the Portugal Wildfire Digital Twin system



Interactive Map

Visualize events on an interactive map with real-time tracking capabilities



Statistical Analysis

Generate AI-driven forecasts and analyze statistical KPIs



Predictive Modeling

LightGBM model for accurate prediction of future wildfire events

[See the System Live](#)

